REQUIREMENT ANALYSIS

Project: SauceDemo (Swag Labs)

Prepared For: QA/Testing & Documentation

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1. Testing Scope

The system under test is a **demo e-commerce application** with core flows:

- Login
- Product browsing
- Cart operations
- Checkout process
- Logout

2. Test Objectives

- Verify all **user workflows**
- Validate **UI behavior**
- Identify **functional defects**
- Ensure **data consistency across pages**

3. User Roles for Testing

Use these standard credentials:

User Type	Purpose
standard_user	Normal flow testing
locked_out_user	Negative login testing
problem_user	UI/functional bugs
performance_glitch_user	Performance delays

4. Functional Areas + Test Requirements

4.1 Login Page Testing

Requirements

- User can login with valid credentials
- Error shown for invalid login
- Locked user cannot login

Manual Test Checks

- Input fields accept text
- Password is masked
- Login button clickable
- Error message visibility & accuracy

Edge Cases

- Empty username/password
- Special characters
- Very long input

4.2 Inventory (Products Page)

Requirements

- Products are displayed correctly
- Sorting works properly

Manual Test Checks

- Product name, price, image visible
- “Add to Cart” button works

- Sorting dropdown functions correctly

Test Scenarios

- Sort A–Z
- Sort Z–A
- Sort by price (low → high, high → low)

4.3 Product Interaction

Requirements

- User can add/remove items

Manual Test Checks

- Button changes:
  - "Add to Cart" → "Remove"
- Cart icon updates count

Edge Cases

- Add multiple items
- Remove all items

4.4 Cart Page

Requirements

- Cart shows correct items
- User can remove items

Manual Test Checks

- Item name, price, quantity
- Remove button works
- Continue shopping button works

4.5 Checkout Process

Step 1: User Info

- Fields required:
  - First Name
  - Last Name
  - ZIP Code

Test Checks

- Empty field validation
- Error messages

Step 2: Overview

- Correct product summary
- Price calculation accurate

Step 3: Completion

- Confirmation message displayed
- “Back Home” button works

4.6 Navigation Menu

Requirements

- Menu options:
  - All Items
  - About
  - Logout
  - Reset App State

Manual Test Checks

- Menu opens/closes

- Navigation works correctly
- Reset clears cart

4.7 Logout

Requirements

- User is logged out successfully

Test Checks

- Redirect to login page
- Session cleared

5. Non-Functional Testing (Manual)

Performance

- Page load time acceptable
- No UI freezing

Usability

- Easy navigation
- Buttons clearly visible
- Consistent layout

Compatibility

Test on:

- Chrome
- Firefox
- Edge

Responsive Testing (Optional)

- Layout on mobile/tablet

6. End-to-End Test Scenario

Scenario: Complete Purchase Flow

1. Login as standard_user
2. Add 2 products
3. Go to cart
4. Checkout
5. Enter details
6. Finish purchase
7. Verify confirmation

7. Negative Testing

- Invalid login
- Empty checkout form
- Remove item not in cart
- Refresh page after adding items

8. Known Testing Areas (Intentional Bugs)

- problem_user → broken images / UI issues
- performance_glitch_user → slow loading

👉 Useful for practicing bug reporting

9. Test Deliverables

- Test scenarios
- Test cases
- Bug reports
- Test execution report

10. Summary

For manual testing, SauceDemo is ideal to practice:

- Functional testing
- UI testing
- Negative testing
- End-to-end scenarios